

# Stephanie J. Spielman, PhD

DATA SCIENTIST

Childhood Cancer Data Lab, Alex's Lemonade Stand Foundation, 3 Bala Plaza, Bala Cynwyd, PA 19004

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*I am a computational biologist and data scientist focused on developing open-source products and educational materials to accelerate pediatric cancer research.*

## Education

### The University of Texas at Austin

PH.D., INTEGRATIVE BIOLOGY

Austin, Texas, USA

2016

### Brown University

B.S., BIOLOGY WITH HONORS

Providence, RI, USA

2010

## Positions

### Alex's Lemonade Stand Foundation

DATA SCIENTIST, CHILDHOOD CANCER DATA LAB

Bala Cynwyd, PA, USA

2022–Present

- Develop open-source tools, data repositories, and training workshops to accelerate pediatric cancer research

### Rowan University

ASSISTANT PROFESSOR

Glassboro, New Jersey, USA

2018–2022

- Ran undergraduate research lab focused on methods in molecular evolution and developed undergraduate-level courses in Evolutionary Biology and Data Science

### Temple University

RESEARCH ASSISTANT PROFESSOR

Philadelphia, Pennsylvania, PA

2016–2018

- Developed research in methods for quantifying protein evolution and taught graduate-level courses in Biostatistics and Evolutionary Medicine

### The University of Texas at Austin

GRADUATE RESEARCH ASSISTANT

Austin, Texas, USA

2011–2016

- Computational molecular evolution research, including phylogenetic modeling and virus evolution.

### The University of Texas at Austin

GRADUATE TEACHING ASSISTANT

Austin, Texas, USA

2011–2016

- Supported undergraduate level courses in Biostatistics, Computational Biology and Bioinformatics, and Evolutionary Biology

## Honors and Awards

### President's Award for Excellence in Innovative Instructional Delivery

ROWAN UNIVERSITY

Glassboro, NJ, USA

2021

- Awarded by the Faculty Center for Teaching Excellence

### Outstanding Dissertation Award

THE UNIVERSITY OF TEXAS AT AUSTIN

Austin, Texas, USA

2016

- Awarded by the Office of Graduate Studies to the top dissertation among all of science, math, and engineering

### Ruth L. Kirschstein National Research Service Award (F31)

THE UNIVERSITY OF TEXAS AT AUSTIN

Austin, Texas, USA

2015

- Pre-doctoral fellowship awarded by NIH/NIGMS

## Selected Publications

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1. Hawkins, A. G., Shapiro, J. A., Spielman, S. J., Mejia, D. S., Prasad, D. V., Ichihara, N., Yakovets, A., Wheeler, K. G., Bethell, C. J., Foltz, S. M., O'Malley, J., Greene, C. S., & Taroni, J. N. (2025). The single-cell pediatric cancer atlas: Data portal and open-source tools for single-cell transcriptomics of pediatric tumors. *bioRxiv*. <https://doi.org/10.1101/2024.04.19.590243>
2. Shapiro, J. A., Gaonkar, K. S., Spielman, S. J., Savonen, C. L., Bethell, C. J., Jin, R., Rathi, K. S., Zhu, Y., Egolf, L. E., Farrow, B. K., Miller, D. P., Yang, Y., Koganti, T., Noureen, N., Koptyra, M. P., Duong, N., Santi, M., Kim, J., Robins, S., ... Taroni, J. N. (2023). OpenPBTA: The open pediatric brain tumor atlas. *Cell Genomics*, 3(7), 100340. <https://doi.org/10.1016/j.xgen.2023.100340>
3. Spielman, S. J., & Miraglia, M. L. (2021). Relative model selection of evolutionary substitution models can be sensitive to multiple sequence alignment uncertainty. *BMC Ecology and Evolution*, 21, 214. <https://doi.org/10.1186/s12862-021-01931-5>
4. Spielman, S. J. (2020). Relative model fit does not predict topological accuracy in single-gene protein phylogenetics. *Molecular Biology and Evolution*, 37(7), 2110–2123. <https://doi.org/10.1093/molbev/msaa075>
5. Spielman, S. J., & Moore, E. K. (2020). dragon: A new tool for exploring redox evolution preserved in the mineral record. *Frontiers in Earth Science*. <https://doi.org/10.3389/feart.2020.585087>
6. Spielman, S. J., & Kosakovsky Pond, S. L. (2018). Relative evolutionary rates in proteins are largely insensitive to the substitution model. *Molecular Biology and Evolution*, 35(9), 2307–2317. <https://doi.org/10.1093/molbev/msy127>
7. Echave, J., Spielman, S. J., & Wilke, C. O. (2016). Causes of evolutionary rate variation among protein sites. *Nature Reviews Genetics*, 17(2), 109–121. <https://doi.org/10.1038/nrg.2015.18>
8. Meyer, A. G., Spielman, S. J., Bedford, T., & Wilke, C. O. (2015). Time dependence of evolutionary metrics during the 2009 pandemic influenza virus outbreak. *Virus Evolution*, 1(1), vev006. <https://doi.org/10.1093/ve/vev006>
9. Spielman, S. J., & Wilke, C. O. (2015). pyvolve: A flexible python module for simulating sequences along phylogenies. *PLOS ONE*, 10(9), e0139047. <https://doi.org/10.1371/journal.pone.0139047>
10. Spielman, S. J., & Wilke, C. O. (2015). The relationship between dN/dS and scaled selection coefficients. *Molecular Biology and Evolution*, 32(4), 1097–1108. <https://doi.org/10.1093/molbev/msv003>